AI-led air superiority. By transforming UAVs into intelligent, decision-capable combat partners, Skyborg is laying the groundwork for the fifth generation of autonomous aerial warfare, where machines will operate seamlessly with humans in complex, multi-domain missions.

Figure 5.5 illustrates the Skyborg Program's architecture, depicting the integration of AI-based autonomy, manned–unmanned teaming (MUM-T), and networked aerial coordination. The diagram highlights how the AI autonomy core functions as the central intelligence, linking a manned aircraft, loyal wingman drones, and autonomous UAVs through a secure data network managed by a ground control station. Arrows indicate the bidirectional exchange of mission data and commands between these platforms, enabling coordinated operations. The inclusion of a UAV swarm demonstrates the system's scalability, allowing multiple drones to collaborate under AI supervision. This figure effectively captures how Skyborg transforms UAVs into intelligent, mission-adaptive partners, capable of real-time learning, distributed decision-making, and cooperative execution alongside human pilots, embodying the future of AI-driven air dominance.